

A Bayesian decision-theoretic approach to multiple testing in basket trials

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Abstract

Basket trials evaluate a single treatment across multiple patient subpopulations, posing important multiple testing problems. Motivated by this setting, we propose a novel Bayesian decision-theoretic approach based on a family of loss functions. These functions adaptively penalize false positives and false negatives based on the numbers of true null and alternative hypotheses, making a basket more likely to be identified as promising when others show promise. Estimation of treatment effects remains independent across baskets, avoiding the computational burden of existing hierarchical modeling approaches and allowing straightforward generalization to trials with different endpoints. The degree of borrowing and the conservativeness of decisions are controlled by tuning parameters, which can be calibrated to achieve desired control of frequentist error rates. The optimal Bayes decision rule is derived by minimizing the posterior expected loss and is computationally efficient. Simulation studies demonstrate that the proposed approach exhibits desirable operating characteristics, performing competitively with widely used alternatives. We illustrate its practical utility through an application to a basket trial of vemurafenib.

Keywords: Bayesian decision theory, borrowing information, clinical trials, family-wise error rate, hierarchical modeling, loss function, multiple testing

1 Introduction

The issue of multiplicity arises when multiple hypotheses are tested simultaneously. This occurs in the analyses of basket trials, in which a single treatment is evaluated across several patient subpopulations (Renfro and Sargent, 2016; Food and Drug Administration, 2022; Woodcock and LaVange, 2017). Here, each subpopulation, referred to as a “basket”, is typically defined by a specific cancer type or subtype in oncology settings. Most basket trials are exploratory, with the goal of identifying baskets with promising response signals for further confirmatory studies (Park et al., 2019). Examples of basket trials include a study of imatinib in multiple histological subtypes of advanced sarcoma (Chugh et al., 2009) and a study of vemurafenib in BRAF V600 mutation-positive non-melanoma cancers (Hyman et al., 2015).

Bayesian approaches to multiple testing typically proceed by assuming a joint prior on the (parameters of the) hypotheses being tested, which introduces an *a priori* dependence structure among the hypotheses (Sjölander and Vansteelandt, 2019). Examples include Berry and Hochberg (1999), Abramovich and Angelini (2006), and Scott and Berger (2010), among others. Bayesian methods of this kind

have gained popularity in the context of basket trials (Pohl et al., 2021; Zhou and Ji, 2024). Since the same treatment is evaluated across all baskets, it is often assumed that the treatment effect in one basket can be informed by treatment effects in other baskets. Through hierarchical modeling, the posterior probabilities of the treatment being efficacious in different baskets are shrunk toward a common value (Thall et al., 2003; Berry et al., 2013; Neuenschwander et al., 2016; Hobbs and Landin, 2018; Chen and Lee, 2020; Zhou and Ji, 2021; Daniells et al., 2023; Pohl et al., 2024; Psioda et al., 2021). By borrowing information across baskets, these methods can increase power in identifying promising baskets compared to analyzing each basket independently. The gain in power may come at the expense of inflated type I error rates when the treatment is efficacious in only a subset of baskets. As a result, recent methodological developments have focused on achieving judicious borrowing to strike a balance between controlling type I error rates and improving statistical power.

Despite the desirable operating characteristics of many hierarchical modeling approaches, they are not without limitations. First, due to the dependence structure of the treatment effects across baskets, posterior inference for these models

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typically relies on computationally intensive methods, such as Markov chain Monte Carlo (MCMC) sampling. Second, most existing hierarchical modeling approaches are developed for non-randomized basket trials with binary endpoints. While generalizations to other types of basket trials (e.g., randomized, continuous endpoints, or mixed endpoints) are possible, they usually require additional model development (Ouma et al., 2022; Zheng and Wason, 2022; Zhao et al., 2023). Third, as noted in Pohl et al. (2024), the complexity of some hierarchical models can make them difficult to communicate to non-statisticians. Sometimes, they involve a large number of hyperparameters, the interpretation and choice of which are not always straightforward. These factors may impede the routine application of hierarchical modeling approaches in basket trials.

The decision-theoretic perspective provides an alternative framework for multiple testing. Through the use of non-additive loss functions, the optimal Bayes decision rule for one hypothesis depends not only on its own posterior probability but also on the posterior probabilities of the other hypotheses. See, for example, Müller et al. (2004), Müller et al. (2006), Cohen et al. (2008), and references therein. To our knowledge, Bayesian decision-theoretic approaches have not yet been applied in the context of basket trials.

For completeness, we note that multiple testing in basket trials has also been studied from a frequentist perspective. Methods such as Cunanan et al. (2017) and He et al. (2022) typically adopt a test-then-pool strategy, where interim futility and homogeneity assessments guide whether baskets should be combined for analysis.

In this paper, we propose a novel Bayesian decision-theoretic approach to multiple testing, motivated by applications in basket trials, that mitigates the limitations of hierarchical modeling approaches. We introduce a family of loss functions that enables information borrowing at the decision level, such that a basket is more likely to be identified as promising when many other baskets show promise, and less likely to be identified when many others do not. Estimation of treatment effects remains independent across baskets, thereby avoiding the computational burden typically associated with hierarchical modeling approaches. The family of loss functions includes tuning parameters that control the degree of borrowing and the conservativeness of decisions, allowing desired control of frequentist error rates. The optimal Bayes decision rule is derived by minimizing the posterior expected loss and, with certain tuning parameters, admits a closed-form expression as simple as classical stepwise procedures. The proposed approach is computationally efficient, conceptually transparent, and readily generalizable to different types of basket trials, including those with mixed endpoint types across baskets. These advantages will become clearer in Section 2, after the necessary notation is introduced. Through simulation, we demonstrate that the proposed approach exhibits desirable operating characteristics (in terms of basket-specific and familywise type I error rate and power) across a wide range of scenarios, performing competitively with some widely used hierarchical modeling approaches.

The remainder of this paper is structured as follows. In Section 2, we describe the proposed approach in detail and estab-

lish some theoretical results. In Section 3, we provide a brief review of existing methods and situate our approach in context. In Section 4, we assess the operating characteristics of the proposed approach through simulation and compare them with existing methods. In Section 5, we apply the proposed approach to a basket trial of vemurafenib. Lastly, in Section 6, we conclude with a discussion. Proofs of the theoretical results are presented in Section S.1 of the [Supplementary Material](#).

2 Proposed method

Consider a basket trial with J baskets. Suppose a total of n_j patients are enrolled in basket j , and let $\mathbf{y}_j = (y_{1j}, \dots, y_{n_jj})'$ denote the data from these patients, where $\mathbf{y}'$ denotes the transpose of $\mathbf{y}$. We assume $\mathbf{y}_j$ has probability density/mass function (pdf/pmf) $f(\mathbf{y}_j | \theta_j)$. The parameter θ_j represents the treatment effect in basket j , which can be a response rate, a mean response, an improvement relative to a control group, or another relevant measure of efficacy. Additional nuisance parameters may be involved in $f(\mathbf{y}_j | \theta_j)$, though they are omitted from the notation for simplicity. It is also allowed that the endpoints are of different types across baskets, in which case the pdf/pmf can be written more generally as $f_j(\mathbf{y}_j | \theta_j)$.

To determine whether the treatment is promising in each basket, the following hypothesis tests are considered,

$$H_{0j} : \theta_j \in \Theta_{0j} \quad \text{versus} \quad H_{1j} : \theta_j \in \bar{\Theta}_{0j}, \quad j = 1, \dots, J.$$

Here, Θ_{0j} is a subset of the parameter space associated with inefficacy of the treatment in basket j , while $\bar{\Theta}_{0j}$ denotes the complement of Θ_{0j} and indicates efficacy. Rejecting H_{0j} means that the treatment is deemed promising and warrants further investigation in that basket.

Most existing Bayesian methods for basket trials proceed by assuming a hierarchical prior on $\theta_1, \dots, \theta_J$ that induces prior and posterior dependence, thereby borrowing information at the treatment effect level. A brief review of these methods will be provided in Section 3.1. In contrast, we assign independent priors to $\theta_1, \dots, \theta_J$, $\theta_j \sim \pi_j(\theta_j)$, which result in independent posterior inference across baskets, $\pi(\theta_1, \dots, \theta_J | \mathbf{y}_1, \dots, \mathbf{y}_J) = \pi_1(\theta_1 | \mathbf{y}_1) \cdots \pi_J(\theta_J | \mathbf{y}_J)$, where $\pi_j(\theta_j | \mathbf{y}_j) \propto \pi_j(\theta_j) f_j(\mathbf{y}_j | \theta_j)$. Conjugate priors are used when possible. This leads to substantially lower computational burden compared to hierarchical modeling approaches.

Although information borrowing does not occur at the treatment effect level, we introduce a family of loss functions that enables borrowing at the decision level. Let v_j be a binary indicator of H_{1j} being true, and let a_j be a binary indicator of the action to reject H_{0j} ,

$$v_j = \begin{cases} 1, & \text{if } \theta_j \in \bar{\Theta}_{0j}; \\ 0, & \text{if } \theta_j \in \Theta_{0j}, \end{cases} \quad \text{and} \quad a_j = \begin{cases} 1, & \text{if } H_{0j} \text{ is rejected;} \\ 0, & \text{if } H_{0j} \text{ is not rejected.} \end{cases} \quad (1)$$

Write $\mathbf{v} = (v_1, \dots, v_J)'$ and $\mathbf{a} = (a_1, \dots, a_J)'$. We propose the following family of loss functions,

$$L(\mathbf{v}, \mathbf{a}; g_1, g_2, c) = c \cdot g_1(J - \mathbf{v}'\mathbf{1}) \cdot \mathbf{a}'(\mathbf{1} - \mathbf{v}) + (1 - c) \cdot g_2(\mathbf{v}'\mathbf{1}) \cdot (\mathbf{1} - \mathbf{a})'\mathbf{v}. \quad (2)$$

Here, $\mathbf{1} = (1, \dots, 1)'$ is a vector of 1's, $(J - \mathbf{v}'\mathbf{1})$ is the number of true null hypotheses, $\mathbf{a}'(\mathbf{1} - \mathbf{v})$ is the number of false

positives, $\mathbf{v}'\mathbf{1}$ is the number of true alternative hypotheses, $(1 - \mathbf{a})'\mathbf{v}$ is the number of false negatives, $g_1, g_2 \geq 0$ are pre-specified nondecreasing functions, and $0 \leq c \leq 1$ is a tuning parameter. The loss function (2) has an intuitive interpretation as increasingly penalizing false positives when many null hypotheses are true, and increasingly penalizing false negatives when many alternative hypotheses are true. This allows for information borrowing at the decision level, as a basket is more likely to be identified as promising when many other baskets show promise, and less likely to be identified when many others do not.

The Bayes decision rule is derived by minimizing the posterior expected loss, as presented in Theorem 1. We write $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_J)$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)$ for simplicity.

Theorem 1: Under loss function (2), the Bayes decision rule that minimizes the posterior expected loss takes the form

$$a_j = \mathbb{I} \left[c \mathbb{E} \{ g_1(J - \mathbf{v}'\mathbf{1})(1 - v_j) | \mathbf{y} \} < (1 - c) \mathbb{E} \{ g_2(\mathbf{v}'\mathbf{1})v_j | \mathbf{y} \} \right], \quad (3)$$

for $j = 1, \dots, J$.

In general, evaluating the expectations in Equation (3) requires numerical integration by simulating from the joint posterior $\pi(\boldsymbol{\theta} | \mathbf{y})$. Let $\{\boldsymbol{\theta}^{(m)} : m = 1, \dots, M\}$ be a sample from $\pi(\boldsymbol{\theta} | \mathbf{y})$, and let $v_j^{(m)} = \mathbb{I}(\theta_j^{(m)} \in \Theta_{0j})$. The expectations, $\mathbb{E}\{g_1(J - \mathbf{v}'\mathbf{1})(1 - v_j) | \mathbf{y}\}$ and $\mathbb{E}\{g_2(\mathbf{v}'\mathbf{1})v_j | \mathbf{y}\}$, can be approximated by $\sum_{m=1}^M g_1(J - \mathbf{v}^{(m)'}\mathbf{1})(1 - v_j^{(m)})/M$ and $\sum_{m=1}^M g_2(\mathbf{v}^{(m)'}\mathbf{1})v_j^{(m)}/M$, respectively.

While Theorem 1 is valid for any joint probability model of $\mathbf{y}$ and $\boldsymbol{\theta}$, we assume prior and posterior independence among $\theta_1, \dots, \theta_J$. This assumption greatly simplifies posterior simulation, as simulating from $\pi(\boldsymbol{\theta} | \mathbf{y})$ reduces to independently sampling from $\pi(\theta_j | \mathbf{y}_j)$ for $j = 1, \dots, J$. Importantly, since the loss function itself induces information borrowing at the decision level, we do not incorporate a dependence structure among the θ_j 's.

2.1 Choice of g_1 and g_2

The functions $g_1(x)$ and $g_2(x)$ are nonnegative, nondecreasing functions on $\{0, \dots, J\}$ that control the degree of borrowing across baskets. Faster-growing functions cause the penalties for false positives and false negatives to increase more rapidly with the numbers of true null and true alternative hypotheses, respectively, thereby inducing stronger borrowing. Empirically, we find that the family of power functions, $g_1(x) = x^{k_1}$ and $g_2(x) = x^{k_2}$ for $k_1, k_2 \geq 0$, provides flexibility across a wide range of scenarios, with $k_1 = k_2 = 0$ implying no borrowing and larger values of k_1 and k_2 yielding stronger borrowing. We will focus on the power function family, although other choices such as $\log(x + 1)$ and e^x are also possible. For brevity, we refer to the proposed method with $g_1(x) = x^{k_1}$ and $g_2(x) = x^{k_2}$ as $L(k_1, k_2)$ in the subsequent discussion. A standard choice is $k_1 = k_2 = 1$, which induces judicious borrowing as shown in simulation studies (Section 4). Stronger borrowing can be achieved by increasing k_1 and/or k_2 , e.g., $k_1 = 1$ and $k_2 = 2$. Conversely, weaker borrowing can be achieved by decreasing k_1 and/or k_2 , e.g., $k_1 = 0$ and $k_2 = 1$. In the extreme case of $k_1 = k_2 = 0$, the loss function

reduces to an additive form, and the proposed method is equivalent to analyzing each basket independently.

For $k_1, k_2 \in \{0, 1\}$, the Bayes decision rule of $L(k_1, k_2)$ has a closed-form expression, the simplicity of which resembles classical stepwise procedures. We present the result for $L(1, 1)$ in Corollary 1.

Corollary 1: Assuming posterior independence of

$v_1, \dots, v_J$. Under loss function (2), for

$g_1(x) = g_2(x) = x$, the Bayes decision rule that

minimizes the posterior expected loss takes the form

$$a_j = \mathbb{I} \left[q_j < 1 - \frac{c \{J - \sum_{l \neq j} (1 - q_l)\}}{c(J + 1) + (1 - 2c) \{1 + \sum_{l \neq j} (1 - q_l)\}} \right], \quad (4)$$

for $j = 1, \dots, J$, where $q_j = \Pr(\theta_j \in \Theta_{0j} | \mathbf{y})$ is the posterior probability of H_{0j} .

Note that posterior independence of $v_1, \dots, v_J$ follows from prior independence and likelihood factorization for $\theta_1, \dots, \theta_J$. Equation (4) shows how the Bayes decision rule for basket j depends on the posterior probabilities of the null hypotheses in other baskets. We present some properties of this decision rule next in Corollary 2.

Corollary 2: Consider the decision rule in Equation (4).

(1) For any $j, j' \in \{1, \dots, J\}$, $q_{j'} \leq q_j$ implies $a_{j'} \geq a_j$.

(2) Let $Q_{-j} = \sum_{l \neq j} q_l \in [0, J - 1]$, and let

$$T_j(Q_{-j}) = 1 - \frac{c(1 + Q_{-j})}{c(J + 1) + (1 - 2c)(J - Q_{-j})} \quad (5)$$

denote the threshold for q_j implied by Equation (4). For $0 < c < 1$, $T_j(Q_{-j})$ is strictly decreasing in Q_{-j} (and for $c = 0$ or 1 , $T_j(Q_{-j}) \equiv 1$ or 0 , respectively).

Part (i) demonstrates that the decision rule is monotonic in q_j , similar to the behavior of classical step-up procedures. Specifically, if $q_{j'} \leq q_j$ and H_{0j} is rejected, then $H_{0j'}$ must also be rejected. Part (ii) characterizes how the rejection threshold for basket j is affected by information from the other baskets. Larger (or smaller) values of Q_{-j} lead to smaller (or larger) T_j , making rejection of H_{0j} more (or less) conservative and reflecting the borrowing mechanism. In particular, T_j is maximized at $1 - c/[J - c(J - 1)]$ when $Q_{-j} = 0$, minimized at $1 - cJ/[1 + c(J - 1)]$ when $Q_{-j} = J - 1$, and equals $1 - c$ when $Q_{-j} = (J - 1)/2$.

Following similar arguments, the Bayes decision rules for $L(1, 0)$ and $L(0, 1)$ are given by

$$a_j = \begin{cases} \mathbb{I} \left[q_j < 1 - \frac{c \{J - \sum_{l \neq j} (1 - q_l)\}}{(cJ + 1) + c \{1 + \sum_{l \neq j} (1 - q_l)\}} \right], & \text{for } L(1, 0); \\ \mathbb{I} \left[q_j < 1 - \frac{c}{c + (1 - c) \{1 + \sum_{l \neq j} (1 - q_l)\}} \right], & \text{for } L(0, 1). \end{cases}$$

Both rules require posterior independence of $v_1, \dots, v_J$ and can be used when weaker borrowing than $L(1, 1)$ is desired. It can be shown that the properties in Corollary 2 continue to hold. For $k_1 = k_2 = 0$, the Bayes decision rule is simply $a_j = \mathbb{I}(q_j < 1 - c)$.

The decision rule for $L(k_1, k_2)$ is generally not symmetric in (k_1, k_2) ; that is, $L(k_1, k_2)$ and $L(k_2, k_1)$ yield different decision rules when $k_1 \neq k_2$. In simulation studies, we observe that for $k_1 > k_2$, $L(k_1, k_2)$ tends to induce stronger borrowing than $L(k_2, k_1)$.

In general, for other choices of g_1 and g_2 , including $g_1(x) = x^{k_1}$ and $g_2(x) = x^{k_2}$ with k_1 or $k_2 \notin \{0, 1\}$, the decision rules follow Equation (3) and do not admit a simpler form due to the loss of linearity in the expectations. Nevertheless, numerical evaluation of Equation (3) remains computationally efficient under posterior independence of $\theta_1, \dots, \theta_J$.

2.2 Choice of c

The tuning parameter c controls the conservativeness of the decision rule. As seen from Equation (3), a_j is a nonincreasing function of c , meaning larger values of c lead to more conservative decisions. The value of c can be calibrated to achieve desired frequentist error control. Given that most basket trials are exploratory, stringent type I error rate control, such as strong control of the family-wise error rate (FWER), is often unnecessary (Wason and Robertson, 2021; Zabor et al., 2022). Still, it may be desirable to guard against false positives, as the incorrect selection of non-promising baskets for further development can lead to increased costs associated with drug development programs that will ultimately fail. We propose calibrating c to achieve weak control of the FWER, that is, controlling the FWER under the global null scenario where the treatment is inefficacious in all baskets. When combined with judicious borrowing induced by appropriate choices of g_1 and g_2 , the FWER is typically maintained within a small margin of the nominal level across all scenarios, including mixed scenarios where the treatment is efficacious in only a subset of baskets.

2.3 Examples and extensions

We provide some examples to illustrate the application of the proposed approach in basket trials. We begin with non-randomized basket trials with binary endpoints, which are the most common type and have been the focus of most methodological developments (Park et al., 2019). In this setting, all n_j patients in basket j receive the investigational treatment. The data $\mathbf{y}_j$ consist of the binary responses from these patients, typically modeled with a Bernoulli distribution,

$$y_{ij} | \theta_j \sim \text{Ber}(\theta_j), \quad i = 1, \dots, n_j, \quad j = 1, \dots, J, \quad (6)$$

where θ_j represents the basket-specific response rate. The goal is to test

$$H_{0j} : \theta_j \leq \theta_{0j} \quad \text{versus} \quad H_{1j} : \theta_j > \theta_{0j}, \quad j = 1, \dots, J, \quad (7)$$

with θ_{0j} being a prespecified reference rate associated with inefficacy for basket j .

We place independent beta priors on θ_j , $\theta_j \sim \text{Beta}(\lambda_j, \eta_j)$, for prespecified λ_j and η_j . For example, the Jeffreys prior corresponds to $\lambda_j = \eta_j = 0.5$. The resulting posterior is $\theta_j | \mathbf{y}_j \sim \text{Beta}(\lambda_j + \sum_i y_{ij}, \eta_j + n_j - \sum_i y_{ij})$. This allows for straightforward and computationally efficient simulation from $\pi(\theta | \mathbf{y})$ and numerical calculation of Equation (3). For $L(k_1, k_2)$ with $k_1, k_2 \in \{0, 1\}$, the Bayes rule depends only on q_j , which can be calculated analytically using the cumulative distribution

function of the beta distribution. This further simplifies the decision rule.

As another example, we consider randomized basket trials with continuous endpoints, as discussed in Zheng and Wason (2022) and Ouma et al. (2022). In this setting, the n_j patients in basket j are randomized to treatment or control. For patient i in basket j , let t_{ij} be a binary indicator of treatment assignment, and y_{ij} the continuous response. Following Zheng and Wason (2022) and Ouma et al. (2022), we model the data using linear regression, $y_{ij} | \beta_j, \theta_j, \sigma_j^2 \sim N(\beta_j + \theta_j t_{ij}, \sigma_j^2)$, where θ_j is the basket-specific treatment effect, and β_j and σ_j^2 are nuisance parameters representing the mean response in the control arm and the error variance in basket j , respectively. The hypothesis test of interest remains as in Equation (7), where θ_{0j} is a minimum clinically important difference. By assigning independent multivariate-normal-inverse-gamma priors to $(\beta_j, \theta_j, \sigma_j^2)$, the resulting posterior is also multivariate-normal-inverse-gamma. As in the previous example, simulation from $\pi(\theta | \mathbf{y})$ and numerical calculation of Equation (3) are straightforward. For $L(k_1, k_2)$ with $k_1, k_2 \in \{0, 1\}$, q_j can be calculated analytically using the cumulative distribution function of the t -distribution. More details are deferred to Section S.2.1 of the [Supplementary Material](#).

The proposed framework is also readily generalizable to trials with mixed endpoint types across baskets. For example, baskets $1, \dots, J_1$ may be non-randomized with binary endpoints, while baskets $J_1 + 1, \dots, J$ may be randomized with continuous endpoints. This setup does not introduce additional complexity to the proposed framework.

In some applications, multiple endpoints are collected for each patient (Wolf et al., 2024; Cao et al., 2025). In this case, the parameter θ_j for basket j becomes a vector, but the overall framework remains unchanged. For example, consider binary efficacy and toxicity outcomes. Let $\theta_j = (\theta_{E,j}, \theta_{T,j})$ denote the efficacy and toxicity rates for basket j . The hypotheses may be formulated as $H_{0j} : \theta_{E,j} \leq \theta_{E,0j}$ or $\theta_{T,j} \geq \theta_{T,0j}$ versus $H_{1j} : \theta_{E,j} > \theta_{E,0j}$ and $\theta_{T,j} < \theta_{T,0j}$, meaning that basket j is considered promising only if sufficient efficacy is achieved without excessive toxicity. Independent priors can again be assigned to θ_j 's across baskets, while within-basket dependence between efficacy and toxicity can be incorporated.

Lastly, in settings where asymmetric borrowing is desirable, such as when baskets differ in variability or clinical importance (see, e.g., Maurer et al., 2023), a more general loss function may be used:

$$L(\mathbf{v}, \mathbf{a}; \mathbf{G}_1, \mathbf{G}_2) = \mathbf{a}' \cdot \mathbf{G}_1(\mathbf{v}) \cdot (1 - \mathbf{v}) + (1 - \mathbf{a})' \cdot \mathbf{G}_2(\mathbf{v}) \cdot \mathbf{v}, \quad (8)$$

where $\mathbf{G}_1(\mathbf{v})$ and $\mathbf{G}_2(\mathbf{v})$ are $J \times J$ (typically diagonal) matrices. The loss function in Equation (2) is a special case of (8) by setting $\mathbf{G}_1(\mathbf{v}) = c \cdot g_1(J - \mathbf{v}'\mathbf{1})\mathbf{I}$ and $\mathbf{G}_2(\mathbf{v}) = (1 - c) \cdot g_2(\mathbf{v}'\mathbf{1})\mathbf{I}$, where $\mathbf{I}$ is the $J \times J$ identity matrix. A flexible specification is obtained by taking $\mathbf{G}_1(\mathbf{v}) = \text{diag}(g_{11}(\mathbf{v}), \dots, g_{1J}(\mathbf{v}))$ and $\mathbf{G}_2(\mathbf{v}) = \text{diag}(g_{21}(\mathbf{v}), \dots, g_{2J}(\mathbf{v}))$, with $g_{1j}(\mathbf{v}) = c_j \left(\sum_{l=1}^J w_{jl}(1 - v_l) \right)^{k_{1j}}$ and $g_{2j}(\mathbf{v}) = (1 - c_j) \left(\sum_{l=1}^J w_{jl}v_l \right)^{k_{2j}}$ for $k_{1j}, k_{2j}, w_{jl} \geq 0$. Un-

der this formulation, k_{1j} and k_{2j} control the overall degree of borrowing for basket j , while w_{jl} controls how much basket j borrows from a specific basket l . See Section S.2.2 of the [Supplementary Material](#) for more details.

3 Existing methods and comparison

3.1 Hierarchical modeling approaches for basket trials

To place our method in context, we provide a brief review of existing methods for basket trials. As noted earlier, most methodological developments have been focused on non-randomized basket trials with binary endpoints; see Equation (6).

The fundamental idea underlying most existing Bayesian methods is to consider a suitable transformation $h(\cdot)$ of θ_j and then model $h(\theta_1), \dots, h(\theta_J)$ as a random sample from a common distribution governed by a vector of hyperparameters ξ ,

$$h(\theta_j) | \xi \stackrel{iid}{\sim} \pi(h(\theta_j) | \xi) \quad \text{for } j = 1, \dots, J, \quad \text{and } \xi \sim \pi(\xi). \quad (9)$$

For example, Thall et al. (2003) apply the transformation $h(\theta_j) = \text{logit}(\theta_j)$ and assume $\pi(h(\theta_j) | \xi)$ is a normal density, with ξ consisting of the unknown normal mean and variance. Berry et al. (2013) consider a modification by adjusting the logit transformations for the target response rates, $h(\theta_j) = \text{logit}(\theta_j) - \text{logit}(\theta_{0j})$. Pohl et al. (2024) work directly with the response rates and assume they follow a beta distribution. A common decision rule is to reject H_{0j} if $\Pr(\theta_j > \theta_{0j} | \mathbf{y}) > \zeta$, where the threshold ζ can vary across baskets if needed.

Building on the pioneering works of Thall et al. (2003) and Berry et al. (2013), subsequent methodological advances have sought to better accommodate heterogeneity in response rates across baskets. For example, Neuenschwander et al. (2016) and Chen and Lee (2020) employ mixture models and latent class models to enable multiple shrinkage. Alternatively, Hobbs and Landin (2018) and Psioda et al. (2021) extend the idea of Bayesian model averaging by considering partitions of baskets into subsets, allowing some response rates to be shared while others differ.

Although the structure of these hierarchical modeling approaches differs from that of the proposed decision-theoretic approach, the underlying philosophies are closely related. For example, in Berry et al. (2013), where $h(\theta_j) = \text{logit}(\theta_j) - \text{logit}(\theta_{0j})$ in Equation (9), shrinking $h(\theta_j)$ toward a common value makes a basket more (or less) likely to be identified as promising when many other baskets exhibit response rates above (or below) the reference rates. This reflects the same core assumption as our proposed method.

As noted in Section 1, these hierarchical modeling approaches are not without limitations. First, due to the prior and posterior dependence of θ_j 's, posterior inference typically relies on MCMC sampling, which can be computationally intensive, especially for large J . Second, these methods may not generalize easily for different data models $f_j(\mathbf{y}_j | \theta_j)$. For example, when the endpoints are of different types across baskets, the unit, scale, or dimension of θ_j may vary, making it inappropriate to assume a common distribution for θ_j 's or

$h(\theta_j)$'s. Third, hierarchical models with improved operating characteristics often involve increased complexity. The choice of the hyperparameters, such as the number and weights of the mixture components in a mixture model or the prior over the space of partitions of baskets, may not be straightforward in practice.

3.2 Other hierarchical modeling approaches

Beyond the basket trials literature, Bayesian hierarchical modeling approaches have been developed for the broader problem of multiple testing. A common strategy is to explicitly model response rates conditional on the hypotheses while specifying a hierarchical prior for the hypotheses,

$$\begin{aligned} \theta_j | v_j &\sim (1 - v_j)\pi_{0j}(\theta_j) + v_j\pi_{1j}(\theta_j) \quad \text{for } j = 1, \dots, J, \\ v_j | \xi_v &\stackrel{iid}{\sim} \pi(v_j | \xi_v) \quad \text{for } j = 1, \dots, J, \quad \text{and } \xi_v \sim \pi(\xi_v). \end{aligned} \quad (10)$$

Here, v_j is a binary indicator of H_{1j} being true (Equation 1), $\pi_{0j}(\theta_j)$ and $\pi_{1j}(\theta_j)$ represent the priors of θ_j under H_{0j} and H_{1j} , respectively, and the hierarchical prior for $v_1, \dots, v_J$ induces dependence among the hypotheses. Such a modeling structure can be found in Abramovich and Angelini (2006), and Scott and Berger (2010). By borrowing information at the hypothesis level rather than at the response rate level, Model (10) avoids the need for the θ_j 's to follow a common distribution, offering greater generalizability compared to Model (9); see Zhou and Ji (2021) for a discussion. In general, posterior inference for Model (10) still requires MCMC sampling. Nevertheless, with certain prior specifications for $v_1, \dots, v_J$, it is possible to derive decision rules that resemble the simplicity of classical stepwise procedures (Abramovich and Angelini, 2006).

3.3 Decision-theoretic approaches

From a Bayesian decision-theoretic perspective, a loss function $L(\mathbf{v}, \mathbf{a})$ is defined, and the Bayes decision rule is obtained by minimizing the posterior expected loss. A common choice is an additive loss function, which sums individual 0-1 losses for each test (Müller et al., 2004), $L_N(\mathbf{v}, \mathbf{a}; c) = c\mathbf{a}'(\mathbf{1} - \mathbf{v}) + (1 - c)(\mathbf{1} - \mathbf{a})'\mathbf{v}$. This is a special case of Equation (2) when $g_1(x) = g_2(x) = 1$. The resulting Bayes decision rule is $a_j = \mathbb{I}(q_j < 1 - c)$, corresponding to the most common rule in basket trial analysis.

Through the use of non-additive loss functions, the Bayes decision rule for one hypothesis depends not only on its own posterior probability but also on those of the other hypotheses. Several alternative loss functions have been explored in Müller et al. (2004; 2006). For example, one loss penalizes false decisions relative to the total numbers of positive and negative decisions, $L_R(\mathbf{v}, \mathbf{a}; c) = c\mathbf{a}'(\mathbf{1} - \mathbf{v})/(\mathbf{a}'\mathbf{1} + \epsilon) + (1 - c)(\mathbf{1} - \mathbf{a})'\mathbf{v}/\{(\mathbf{1} - \mathbf{a})'\mathbf{1} + \epsilon\}$, where the additional term ϵ avoids a zero denominator. Cohen et al. (2008) first propose the idea of increasingly penalizing false negatives when many alternative hypotheses are true, leading to $L_C(\mathbf{v}, \mathbf{a}; c) = c\mathbf{a}'(\mathbf{1} - \mathbf{v}) + (1 - c)(\mathbf{v}'\mathbf{1})(\mathbf{1} - \mathbf{a})'\mathbf{v}$. This is a special case of Equation (2) when $g_1(x) = 1$ and $g_2(x) = x$.

It is important to note that the hierarchical modeling and decision-theoretic frameworks are not mutually exclusive. Rather, these two frameworks complement each other and can be used in conjunction to achieve desired statistical properties (Berry and Hochberg, 1999; Müller et al., 2006). Most existing Bayesian methods for basket trials pair a complicated hierarchical model for θ with a simple decision rule. In contrast, our approach takes the opposite strategy, combining a simple model for θ with a tailored decision rule. This allows for greater computational efficiency and improved generalizability and interpretability.

4 Simulation studies

We assess the operating characteristics of the proposed decision-theoretic approach through a series of simulation studies. The method $L(k_1, k_2)$ is implemented for several combinations of (k_1, k_2) . We present results for $(k_1, k_2) = (1, 1)$ and $(1, 2)$, with additional results for $(0, 0)$, $(0, 1)$, and $(2, 1)$ reported in the [Supplementary Material](#). The performance metrics considered include: (i) the basket-specific type I error rate, defined as the probability of incorrectly rejecting H_{0j} for a non-promising basket j , and the basket-specific power, defined as the probability of correctly rejecting H_{0j} for a promising basket j ; (ii) the FWER, defined as the probability of incorrectly rejecting at least one true null hypothesis across all baskets; and (iii) the family-wise power (FWP), where two notions are considered: the disjunctive power (FWP-D), defined as the probability of correctly rejecting at least one false null hypothesis across all baskets, and the conjunctive power (FWP-C), defined as the probability of correctly rejecting all false null hypotheses. These probabilities are approximated by the relative frequencies of the corresponding events over a large number of simulated trials.

We begin by considering non-randomized basket trials with binary endpoints. Motivated by the vemurafenib trial (Hyman et al., 2015; Daniells et al., 2023), we consider $J = 5$ baskets with a common reference response rate $\theta_{0j} = 0.15$. We examine the response-rate scenarios in Daniells et al. (2023) under two sample size settings: (i) planned sample sizes with 13 patients per basket, and (ii) realized sample sizes from the vemurafenib trial, (20, 10, 8, 18, 7). Scenarios 1–5 contain 1–5 promising baskets, respectively. In promising baskets, the response rate is set to 0.45, while in non-promising baskets, it remains at the reference rate of 0.15. Scenarios 6–9 involve weaker and heterogeneous signals, with some promising baskets having a response rate of 0.35. See Tables 2–5. The binary responses in each basket are generated from a Bernoulli distribution (Equation 6). For each scenario, we simulate 2000 trials.

To implement the proposed method, we place independent Beta(0.5, 0.5) priors on θ_j . For comparison, we also implement the following methods. (i) The Bonferroni correction, implemented by conducting Fisher's exact test within each basket at a significance level of 0.02, yielding strong control of the FWER at the 10% level. (ii) A Bayesian hierarchical model (BHM) applied to the logit transformed response rates with an

Table 1 Tuning parameters and posterior thresholds used to achieve weak FWER control at the 10% level in the 5-basket setting, under planned and realized sample sizes.

Method	Param	Planned	Realized
$L(1, 1)$	c	0.9831	0.9804
$L(1, 2)$	c	0.9946	0.9943
BHM	ζ	0.9348	0.9355
EXNEX(0.25, 0.25)	ζ	0.9642	0.9568
EXNEX(0.375, 0.375)	ζ	0.9545	0.9513
MEM	ζ	0.9555	0.9560

adjustment for the target response rate, $\text{logit}(\theta_j) - \text{logit}(\theta_{0j}) \mid \mu, \tau^2 \sim N(\mu, \tau^2)$, $\mu \sim N(0, 10^2)$, and $\tau \sim \text{Half-Cauchy}(25)$. See Thall et al. (2003), Berry et al. (2013), and Daniells et al. (2023). (iii) The exchangeability-nonexchangeability model (EXNEX; Neuenschwander et al., 2016),

$$\text{logit}(\theta_j) - \text{logit}(\theta_{0j}) \mid \mathbf{w}, \mu, \tau^2 \sim \sum_{d=1}^D w_d N(\mu_{\text{EX},d}, \tau_{\text{EX},d}^2) + w_0 N(\mu_{\text{NEX},j}, \tau_{\text{NEX},j}^2).$$

We use EXNEX($w_1, \dots, w_D$) to denote an EXNEX model that has D EX components with weights ($w_1, \dots, w_D$) and 1 NEX component with weight $w_0 = 1 - \sum_{d=1}^D w_d$. We present results for EXNEX(0.25, 0.25) and EXNEX(0.375, 0.375), with additional results for EXNEX(0.25), EXNEX(0.5), and EXNEX(0.75) deferred to the [Supplementary Material](#). Hyperpriors follow Neuenschwander et al. (2016) and are detailed in Section S.3.1 of the Supplementary Material. (iv) The symmetric multisource exchangeability model (MEM; Hobbs and Landin, 2018), which is implemented via the R package `basket` (Kane et al., 2021) with the default hyperparameter setting. For BHM, EXNEX, and MEM, H_{0j} is rejected when $\Pr(\theta_j > \theta_{0j} \mid \mathbf{y}) > \zeta$. To calibrate the tuning parameter c in $L(k_1, k_2)$ and the threshold ζ , we consider an additional global null scenario, in which the treatment response rate is 0.15 in all 5 baskets. We simulate 5000 trials under this scenario and choose c and ζ to achieve weak control of the FWER at the 10% level. The resulting values of c and ζ are reported in Table 1. For Bayesian methods that require Monte Carlo sampling, 5000 posterior draws are taken (following 1000 burn-in draws where applicable).

Tables 2 and 3 summarize the simulation scenarios and results under the planned sample size setting. Methods that borrow information across baskets achieve higher power than Bonferroni, which does not borrow information. BHM and MEM exhibit inflated type I error rates in the mixed scenarios, which may raise concerns for trial sponsors (worst-case FWERs of 22.4% for BHM in Scenario 4 and 42.3% for MEM in Scenario 7). Both $L(k_1, k_2)$ and EXNEX allow straightforward tuning of the degree of borrowing. Stronger borrowing generally increases both power and type I error, reflecting the inherent trade-off between these two quantities. For $L(1, 1)$, $L(1, 2)$, EXNEX(0.25, 0.25), and EXNEX(0.375, 0.375), type I error control is reasonable, with FWER maintained within a small margin above the nominal 10% level. The worst-

Table 2 Simulation results (Scenarios 1–5) for non-randomized basket trials with binary endpoints across 5 baskets, based on a planned sample size of 13 patients per basket.

Method	% Reject					FWER	FWP-D	FWP-C
Scenario 1: $\theta = (0.45, 0.15, 0.15, 0.15, 0.15)$								
$L(1, 1)$	67.5	2.9	3.3	3.3	2.5	11.1	67.5	67.5
$L(1, 2)$	67.5	3.1	3.5	3.3	2.8	11.4	67.5	67.5
Bonferroni	57.2	0.8	0.7	0.9	0.7	2.9	57.2	57.2
BHM	68.7	5.7	6.2	6.0	5.2	17.9	68.7	68.7
EXNEX(0.25, 0.25)	69.1	3.6	3.8	3.6	3.0	13.2	69.1	69.1
EXNEX(0.375, 0.375)	68.1	3.6	3.8	3.5	3.2	13.2	68.1	68.1
MEM	68.0	10.6	10.2	10.6	10.0	20.0	68.0	68.0
Scenario 2: $\theta = (0.45, 0.45, 0.15, 0.15, 0.15)$								
$L(1, 1)$	72.8	74.1	4.0	3.3	3.6	9.8	92.2	54.7
$L(1, 2)$	72.7	74.3	4.7	4.4	4.2	11.1	91.8	55.2
Bonferroni	56.2	58.7	1.0	1.0	1.1	3.1	81.9	33.0
BHM	79.1	80.1	9.4	8.7	9.1	21.7	93.4	65.7
EXNEX(0.25, 0.25)	74.5	76.9	3.9	3.3	3.4	10.0	92.8	58.6
EXNEX(0.375, 0.375)	75.2	77.4	4.1	3.8	3.6	10.3	93.1	59.5
MEM	77.2	81.3	20.1	18.6	20.3	33.0	91.5	67.0
Scenario 3: $\theta = (0.45, 0.45, 0.45, 0.15, 0.15)$								
$L(1, 1)$	77.1	77.3	77.6	5.0	4.6	8.3	98.4	46.9
$L(1, 2)$	79.0	79.3	79.1	7.6	7.6	13.5	98.5	51.1
Bonferroni	58.4	58.0	59.1	0.9	0.8	1.6	91.9	20.8
BHM	84.8	86.1	85.0	14.4	14.4	22.1	98.8	64.9
EXNEX(0.25,0.25)	76.9	77.4	77.9	4.0	3.3	7.1	98.5	46.7
EXNEX(0.375,0.375)	78.1	78.7	78.6	6.0	5.5	10.0	98.4	49.1
MEM	90.9	90.0	89.4	30.4	28.3	41.6	98.6	78.1
Scenario 4: $\theta = (0.45, 0.45, 0.45, 0.45, 0.15)$								
$L(1, 1)$	78.0	76.6	78.9	78.5	8.0	8.0	99.8	38.9
$L(1, 2)$	82.4	81.5	83.0	83.0	10.5	10.5	99.8	51.5
Bonferroni	58.3	56.0	57.7	57.2	0.1	0.1	96.6	10.5
BHM	89.7	89.1	89.6	90.1	22.4	22.4	99.9	67.3
EXNEX(0.25,0.25)	76.6	75.1	77.7	77.1	2.7	2.7	99.8	34.4
EXNEX(0.375,0.375)	78.9	78.3	80.0	79.8	9.1	9.1	99.8	42.4
MEM	95.0	94.5	94.3	94.7	35.7	35.7	99.9	83.9
Scenario 5: $\theta = (0.45, 0.45, 0.45, 0.45, 0.45)$								
$L(1, 1)$	86.4	86.2	86.1	85.1	86.7	0.0	99.9	57.7
$L(1, 2)$	89.8	89.9	89.9	89.3	90.5	0.0	100.0	60.7
Bonferroni	57.3	56.6	58.6	57.2	58.0	0.0	98.7	6.8
BHM	95.1	94.9	94.2	95.2	95.2	0.0	100.0	80.2
EXNEX(0.25,0.25)	76.5	77.8	77.9	77.0	77.5	0.0	99.9	27.9
EXNEX(0.375,0.375)	87.2	87.1	87.4	86.4	87.3	0.0	99.9	57.5
MEM	97.4	97.4	96.9	97.2	97.6	0.0	100.0	88.2

Notes: The 5 columns under “% Reject” report the basket-specific type I error rate (non-bold, for non-promising baskets) and power (bold, for promising baskets). The FWER under the global null scenario is controlled at 10% for all methods.

case FWERs are 11.1% for $L(1, 1)$ (Scenario 1), 13.5% for $L(1, 2)$ (Scenario 3), 13.2% for EXNEX(0.25, 0.25) (Scenario 1), and 13.2% for EXNEX(0.375, 0.375) (Scenario 1). Under these specifications, $L(k_1, k_2)$ and EXNEX exhibit overall comparable performance, although scenario-specific differences remain. From a computational perspective, $L(k_1, k_2)$ does not require intensive MCMC sampling and is therefore substantially faster than BHM, EXNEX, and MEM, which may be advantageous in practice. Tables 4 and 5 report results under

the realized sample-size setting and show similar performance trends.

Section S.3.1 of the [Supplementary Material](#) presents the complete simulation results, including results for $L(0, 0)$, $L(0, 1)$, $L(2, 1)$, EXNEX(0.75), EXNEX(0.5), and EXNEX(0.25), as well as additional scenarios under the realized sample size setting. We also report sensitivity analyses of $L(k_1, k_2)$ under alternative priors for θ_j , including Beta(0.001, 0.001) and Beta(1, 1).

Table 3 Simulation results (Scenarios 6–9) for non-randomized basket trials with binary endpoints across 5 baskets, based on a planned sample size of 13 patients per basket.

Method	% Reject					FWER	FWP-D	FWP-C
Scenario 6: $\theta = (0.35, 0.15, 0.15, 0.15, 0.15)$								
$L(1, 1)$	41.2	3.0	2.5	2.8	2.2	9.7	41.2	41.2
$L(1, 2)$	41.2	3.1	2.5	3.0	2.4	9.9	41.2	41.2
Bonferroni	29.2	0.8	1.1	0.5	0.8	3.0	29.2	29.2
BHM	42.7	5.0	4.7	4.7	3.9	14.3	42.7	42.7
EXNEX(0.25,0.25)	42.1	3.2	2.8	3.2	2.4	10.9	42.1	42.1
EXNEX(0.375,0.375)	42.7	3.3	2.9	3.4	2.6	11.4	42.7	42.7
MEM	42.7	9.3	10.3	9.9	9.5	17.7	42.7	42.7
Scenario 7: $\theta = (0.35, 0.35, 0.35, 0.15, 0.15)$								
$L(1, 1)$	50.3	51.0	47.1	4.2	4.2	7.8	85.4	12.8
$L(1, 2)$	51.8	52.4	49.1	5.7	6.1	10.8	85.2	15.7
Bonferroni	30.5	29.3	27.5	1.1	0.9	2.0	64.0	2.6
BHM	61.2	62.3	59.7	13.0	13.3	20.9	87.5	30.5
EXNEX(0.25,0.25)	50.5	52.0	47.5	3.8	3.8	7.4	85.5	12.8
EXNEX(0.375,0.375)	51.4	52.9	48.3	4.7	5.3	9.1	85.8	13.8
MEM	71.8	71.8	70.4	31.7	33.0	42.3	87.9	54.5
Scenario 8: $\theta = (0.45, 0.35, 0.35, 0.15, 0.15)$								
$L(1, 1)$	76.7	48.1	48.6	4.2	3.0	6.5	93.2	18.5
$L(1, 2)$	78.2	50.4	51.0	6.8	5.4	10.9	93.4	22.5
Bonferroni	59.1	28.5	27.4	1.2	0.4	1.5	78.9	4.8
BHM	83.4	60.9	61.6	13.6	13.0	20.5	94.2	37.3
EXNEX(0.25,0.25)	77.2	48.4	48.9	3.6	2.5	5.9	93.0	18.6
EXNEX(0.375,0.375)	78.0	49.1	50.3	5.1	3.6	7.8	93.5	20.3
MEM	87.5	71.4	70.6	31.4	30.9	41.8	93.6	57.8
Scenario 9: $\theta = (0.45, 0.45, 0.35, 0.35, 0.15)$								
$L(1, 1)$	77.1	78.2	52.1	51.5	6.4	6.4	98.4	18.2
$L(1, 2)$	81.2	81.8	58.5	58.2	9.2	9.2	98.4	29.1
Bonferroni	58.9	57.7	29.3	28.9	1.0	1.0	91.6	2.5
BHM	88.5	88.3	70.1	70.9	21.3	21.3	99.1	45.5
EXNEX(0.25,0.25)	76.5	77.7	50.4	49.8	3.1	3.1	98.5	14.8
EXNEX(0.375,0.375)	78.5	79.4	54.0	53.8	7.6	7.6	98.4	21.7
MEM	94.7	94.4	82.6	83.3	40.3	40.3	99.4	71.0

Notes: The 5 columns under “% Reject” report the basket-specific type I error rate (non-bold, for non-promising baskets) and power (bold, for promising baskets). The FWER under the global null scenario is controlled at 10% for all methods.

For additional validation, we extend the simulations to a 10-basket setting; see Section S.3.2 of the [Supplementary Material](#). To further demonstrate the versatility of the proposed method, we also conduct simulations for randomized basket trials with continuous endpoints and compare with the methods of Zheng and Wason (2022) and Ouma et al. (2022); see Section S.3.3 of the [Supplementary Material](#).

5 Case study

We apply the proposed approach to a non-randomized trial of vemurafenib in patients with BRAF V600 mutation-positive non-melanoma cancers (Hyman et al., 2015). The primary endpoint was binary tumor response, evaluated at week 8 after treatment. The original design and analysis were conducted separately for each cancer cohort, and the data were subsequently reanalyzed by Daniells et al. (2023). Following Daniells et al. (2023), we focus on five baskets in our anal-

ysis: non-small-cell lung cancer (NSCLC), colorectal cancer (CRC), cholangiocarcinoma (CCA), Erdheim–Chester disease or Langerhans cell histiocytosis (ECD/LCH), and anaplastic thyroid cancer (ATC). A response rate below $\theta_{0j} = 15\%$ was considered indicative of inefficacy. A total of 63 patients are available for analysis, and the sample size and number of responses per basket are shown in the second column of Table 6.

We model the binary responses using the Bernoulli model in Equation (6). To apply $L(k_1, k_2)$, we place independent $\text{Beta}(0.5, 0.5)$ priors on θ_j , yielding the posterior $\text{Beta}(0.5 + \sum_i y_{ij}, 0.5 + n_j - \sum_i y_{ij})$. The posterior means and 95% equal-tailed credible intervals for θ_j are reported in the top panel of Table 6. Note that information borrowing does not occur at the treatment effect level, and these estimates are identical to those from independent Bayesian analyses of each basket. Since decisions under $L(k_1, k_2)$ are based on minimizing posterior expected loss, they need not coincide with whether the null value θ_{0j} lies inside the credible interval.

Table 4 Simulation results (Scenarios 1–5) for non-randomized basket trials with binary endpoints across 5 baskets, based on realized sample sizes of (20, 10, 8, 18, 7).

Method	% Reject					FWER	FWP-D	FWP-C
Scenario 1: $\theta = (0.45, 0.15, 0.15, 0.15, 0.15)$								
$L(1, 1)$	86.7	3.7	2.9	3.3	3.1	11.9	86.7	86.7
$L(1, 2)$	85.1	3.7	3.3	3.5	3.4	12.5	85.1	85.1
Bonferroni	75.0	0.8	0.4	1.3	1.2	3.6	75.0	75.0
BHM	88.2	6.5	5.2	7.1	5.6	19.8	88.2	88.2
EXNEX(0.25,0.25)	87.7	4.4	2.6	4.4	1.7	12.3	87.7	87.7
EXNEX(0.375,0.375)	88.2	4.6	2.9	4.5	2.9	13.6	88.2	88.2
MEM	85.9	12.2	11.4	12.0	11.9	24.3	85.9	85.9
Scenario 2: $\theta = (0.45, 0.45, 0.15, 0.15, 0.15)$								
$L(1, 1)$	86.8	66.5	3.8	4.1	5.8	11.9	95.3	58.0
$L(1, 2)$	87.3	67.1	5.1	4.5	6.4	13.7	94.9	59.6
Bonferroni	73.8	50.2	0.5	1.4	1.7	3.4	86.7	37.3
BHM	91.5	74.6	9.8	8.8	9.1	21.5	95.9	70.2
EXNEX(0.25,0.25)	86.5	69.5	2.2	4.0	4.0	9.4	94.8	61.2
EXNEX(0.375,0.375)	88.7	71.4	4.3	4.3	6.7	13.3	95.2	64.9
MEM	90.8	77.7	22.1	16.7	22.5	34.7	95.5	73.0
Scenario 3: $\theta = (0.45, 0.45, 0.45, 0.15, 0.15)$								
$L(1, 1)$	89.4	71.2	58.1	5.2	6.6	11.1	98.0	40.0
$L(1, 2)$	90.9	71.1	62.2	7.4	6.6	13.2	97.9	44.1
Bonferroni	75.3	50.5	27.2	1.2	1.1	2.3	90.7	10.7
BHM	94.7	78.3	72.4	13.6	12.5	20.7	98.7	57.3
EXNEX(0.25,0.25)	88.4	72.8	54.3	4.4	5.8	9.9	97.7	37.5
EXNEX(0.375,0.375)	92.0	73.4	60.1	5.6	6.7	11.6	98.2	43.8
MEM	94.6	86.1	82.2	24.2	33.6	41.5	98.5	72.9
Scenario 4: $\theta = (0.45, 0.45, 0.45, 0.45, 0.15)$								
$L(1, 1)$	92.8	71.3	68.0	90.3	6.2	6.2	99.9	43.4
$L(1, 2)$	93.6	73.9	71.2	91.5	6.2	6.2	99.9	48.6
Bonferroni	74.7	48.1	27.1	77.4	0.5	0.5	98.1	8.1
BHM	96.4	84.0	82.3	95.0	24.0	24.0	100.0	66.4
EXNEX(0.25,0.25)	92.1	71.2	54.2	89.3	6.1	6.1	99.9	33.3
EXNEX(0.375,0.375)	94.0	71.4	75.1	90.5	6.2	6.2	100.0	48.4
MEM	97.7	93.4	91.4	96.8	48.7	48.7	100.0	83.9
Scenario 5: $\theta = (0.45, 0.45, 0.45, 0.45, 0.45)$								
$L(1, 1)$	94.2	74.0	76.6	92.3	68.6	0.0	99.9	36.3
$L(1, 2)$	95.4	83.8	78.0	94.1	68.6	0.0	100.0	44.4
Bonferroni	73.5	49.1	27.3	77.1	39.7	0.0	98.7	3.0
BHM	97.6	90.2	89.1	96.9	87.1	0.0	100.0	71.3
EXNEX(0.25,0.25)	93.4	73.8	59.5	88.5	68.0	0.0	100.0	28.1
EXNEX(0.375,0.375)	94.2	73.9	77.7	91.7	68.6	0.0	100.0	35.4
MEM	98.2	95.9	94.0	97.8	94.7	0.0	100.0	84.1

Notes: The 5 columns under “% Reject” report the basket-specific type I error rate (non-bold, for non-promising baskets) and power (bold, for promising baskets). The FWER under the global null scenario is controlled at 10% for all methods.

To identify which baskets are promising, we consider $L(1, 1)$ and $L(1, 2)$. The tuning parameter c is taken from the 5-basket simulation setting with realized sample sizes (Table 1), which ensures weak control of the FWER at the 10% level when the response rate is 0.15 in all baskets. Specifically, $c_{L(1,1)} = 0.9804$ and $c_{L(1,2)} = 0.9943$. Under $L(1, 1)$, the Bayes rule admits a closed-form expression. We first calculate $q_j = \Pr(\theta_j \leq \theta_{0j} | \mathbf{y}_j)$ using the cumulative distribution function of the Beta posterior; the resulting values are reported in the bottom panel of Table 6. Next, for each basket j ,

we calculate the threshold T_j according to Equation (5). For NSCLC, CRC, CCA, ECD/LCH, and ATC, the thresholds are 0.025, 0.048, 0.036, 0.025, and 0.027, respectively. Based on whether $q_j < T_j$, the NSCLC and ECD/LCH baskets are identified as promising. Under $L(1, 2)$, the Bayes rule has to be evaluated numerically. We draw $\{\theta_j^{(m)}, m = 1, \dots, M\}$ from the beta posterior for $M = 5000$, calculate $v_j^{(m)} = \mathbb{I}(\theta_j^{(m)} > \theta_{0j})$, and approximate the posterior expectations in Equation (3) by Monte Carlo integration. This procedure again identifies the NSCLC and ECD/LCH baskets as promising.

Table 5 Simulation results (Scenarios 6–9) for non-randomized basket trials with binary endpoints across 5 baskets, based on realized sample sizes of (20, 10, 8, 18, 7).

Method	% Reject					FWER	FWP-D	FWP-C
Scenario 6: $\theta = (0.35, 0.15, 0.15, 0.15, 0.15)$								
$L(1, 1)$	57.5	3.4	2.3	3.1	2.4	9.9	57.5	57.5
$L(1, 2)$	55.6	3.2	3.0	3.0	2.8	10.4	55.6	55.6
Bonferroni	40.3	1.1	0.4	1.0	1.0	3.3	40.3	40.3
BHM	60.9	5.9	4.6	6.1	3.8	16.4	60.9	60.9
EXNEX(0.25,0.25)	59.3	4.3	2.1	4.3	1.3	11.1	59.3	59.3
EXNEX(0.375,0.375)	60.6	4.3	2.5	4.1	2.4	11.8	60.6	60.6
MEM	58.1	12.3	13.2	13.0	12.5	23.8	58.1	58.1
Scenario 7: $\theta = (0.35, 0.35, 0.35, 0.15, 0.15)$								
$L(1, 1)$	63.6	47.3	32.1	4.9	6.6	11.1	84.8	11.9
$L(1, 2)$	65.1	46.9	35.8	6.0	6.9	12.3	85.1	13.9
Bonferroni	41.8	26.6	10.3	1.6	1.1	2.6	60.9	1.1
BHM	73.3	55.2	45.5	12.3	11.2	19.1	86.4	25.6
EXNEX(0.25,0.25)	61.8	47.1	28.4	4.8	4.0	8.5	82.6	8.8
EXNEX(0.375,0.375)	67.4	48.6	32.8	5.3	5.8	10.6	84.9	13.7
MEM	75.3	66.2	61.3	26.9	33.9	40.5	85.8	48.4
Scenario 8: $\theta = (0.45, 0.35, 0.35, 0.15, 0.15)$								
$L(1, 1)$	89.2	44.3	32.9	5.2	5.8	10.2	94.9	14.2
$L(1, 2)$	89.5	44.5	37.0	6.6	6.0	11.7	94.9	17.0
Bonferroni	75.4	24.4	10.6	1.5	0.7	2.2	83.1	1.8
BHM	92.4	53.2	46.7	12.3	11.0	18.6	95.9	28.0
EXNEX(0.25,0.25)	88.2	45.5	28.3	4.4	3.8	7.8	94.2	11.8
EXNEX(0.375,0.375)	90.9	46.2	33.6	5.4	5.8	10.5	95.4	16.2
MEM	92.5	63.9	61.8	23.7	28.9	36.0	95.7	48.5
Scenario 9: $\theta = (0.45, 0.45, 0.35, 0.35, 0.15)$								
$L(1, 1)$	91.4	73.6	44.7	64.9	6.6	6.6	99.0	22.3
$L(1, 2)$	92.7	75.2	49.2	68.6	6.7	6.7	99.3	28.1
Bonferroni	73.8	49.6	10.2	45.1	1.4	1.4	93.0	1.8
BHM	96.3	83.4	62.8	79.0	22.8	22.8	99.7	45.8
EXNEX(0.25,0.25)	90.9	73.6	31.5	63.7	6.3	6.3	99.0	13.6
EXNEX(0.375,0.375)	94.0	73.9	50.8	65.8	6.6	6.6	99.4	27.1
MEM	98.2	93.7	81.5	87.1	51.4	51.4	99.8	72.1

Notes: The 5 columns under “% Reject” report the basket-specific type I error rate (non-bold, for non-promising baskets) and power (bold, for promising baskets). The FWER under the global null scenario is controlled at 10% for all methods.

Table 6 Analysis of the vemurafenib trial. The column y/n reports the number of responses/sample size. Under each method, $\hat{\theta}$ is a point estimate of the response rate (the posterior mean for Bayesian methods), “95% CI” denotes a 95% credible interval or confidence interval, q denotes the posterior probability of the null, a indicates whether the null is rejected ($a = 1$) or not ($a = 0$), and p is the p -value from Fisher’s exact test. The reference response rate is $\theta_{0j} = 0.15$ for all baskets.

Basket	y/n	$L(k_1, k_2)$			EXNEX(0.25,0.25)		EXNEX(0.375,0.375)		Bonferroni	
Inference		$\hat{\theta}$ (95% CI)			$\hat{\theta}$ (95% CI)		$\hat{\theta}$ (95% CI)		$\hat{\theta}$ (95% CI)	
1. NSCLC	8/20	0.405 (0.211, 0.616)			0.382 (0.202, 0.583)		0.373 (0.202, 0.575)		0.400 (0.191, 0.639)	
2. CRC	0/10	0.045 (0.000, 0.217)			0.044 (0.000, 0.210)		0.054 (0.000, 0.247)		0.000 (0.000, 0.308)	
3. CCA	1/8	0.167 (0.014, 0.454)			0.153 (0.010, 0.428)		0.168 (0.012, 0.420)		0.125 (0.003, 0.527)	
4. ECD/LCH	6/18	0.342 (0.153, 0.563)			0.323 (0.140, 0.528)		0.327 (0.150, 0.530)		0.333 (0.133, 0.590)	
5. ATC	2/7	0.312 (0.065, 0.648)			0.283 (0.052, 0.607)		0.291 (0.053, 0.571)		0.286 (0.037, 0.710)	
Decision		q	$a_{L(1,1)}$	$a_{L(1,2)}$	q	a	q	a	p	a
1. NSCLC	8/20	0.003	1	1	0.003	1	0.003	1	0.006	1
2. CRC	0/10	0.932	0	0	0.936	0	0.901	0	1.000	0
3. CCA	1/8	0.528	0	0	0.565	0	0.509	0	0.728	0
4. ECD/LCH	6/18	0.023	1	1	0.035	1	0.025	1	0.042	0
5. ATC	2/7	0.153	0	0	0.190	0	0.160	0	0.283	0

For comparison, we also report point and interval estimates of θ_j from independent frequentist analyses of each basket (maximum likelihood estimates with Clopper–Pearson intervals), as well as results from EXNEX(0.25,0.25) and EXNEX(0.375,0.375); see the top panel of Table 6. For decision making under Bonferroni correction, Fisher’s exact test is conducted within each basket at a significance level of 0.02, under which only the NSCLC basket is deemed promising. For EXNEX, we compute q_j and apply the corresponding thresholds in Table 1. Specifically, $\zeta_{\text{EXNEX}(0.25,0.25)} = 0.9568$ and $\zeta_{\text{EXNEX}(0.375,0.375)} = 0.9513$. Under both configurations, based on whether $q_j < 1 - \zeta$, the NSCLC and ECD/LCH baskets are identified as promising. To summarize, $L(1, 1)$, $L(1, 2)$, and EXNEX identify one additional promising basket relative to Bonferroni, illustrating the potential efficiency gains from information borrowing.

For additional illustration, we also apply the proposed approach to a non-randomized basket trial of imatinib with binary endpoints (Chugh et al., 2009) and a randomized trial of a dietary intervention with continuous endpoints (Ouma et al., 2022). The results are provided in Section S.4 of the [Supplementary Material](#).

6 Discussion

We have proposed a Bayesian decision-theoretic approach to multiple testing in basket trials, formulated through a flexible family of loss functions. The proposed method is computationally efficient, conceptually transparent, and easily generalizable to different types of basket trials. Although the method is motivated by basket trials, it is broadly applicable to other multiple testing problems such as variable selection, where the hypotheses take the form $H_{0j} : \theta_j = 0$ vs $H_{1j} : \theta_j \neq 0$.

We have focused on the power function family for g_1 and g_2 . It may be worthwhile to explore alternative forms for g_1 and g_2 , such as exponential or piecewise functions. We have proposed to calibrate the tuning parameter c to control the conservativeness of the decision rule and achieve desired frequentist error control. Alternatively, one can fix c and instead adjust the prior probability assigned to each null hypothesis. We have focused on weak control of the FWER. Alternatively, one may consider controlling the basket-specific type I error rate under the global null, or an average type I error rate across multiple scenarios (Daniells et al., 2025). We have employed conjugate priors for each θ_j , which are a computationally efficient choice. To further improve operating characteristics, alternative priors such as non-local priors can be considered (Johnson and Rossell, 2010).

As discussed in Section 3.1, the underlying philosophies of decision-theoretic and hierarchical modeling approaches are closely related. Further exploration of their connections remains an interesting direction for future research. The choices of the loss components g_1 , g_2 , and c are primarily motivated by achieving desirable frequentist operating characteristics. A potential avenue for future work is to explore their specifications that reflect a substantive decision-theoretic loss.

It is important to note that the proposed family of loss functions can be used in conjunction with any joint probability

model of $\mathbf{y}$ and $\boldsymbol{\theta}$ to derive an optimal decision rule. Since the loss functions inherently induce information borrowing at the decision level, we intentionally avoided introducing a dependence structure among the θ_j ’s, which could otherwise result in over-borrowing. Estimation of treatment effects remains independent across baskets, which presents both advantages and limitations. On one hand, the independence substantially improves computational efficiency and helps avoid potential bias in estimating each θ_j . On the other hand, not incorporating a hierarchical structure among the θ_j ’s precludes potential gains in estimation efficiency, such as a reduction in mean squared error.

Throughout this paper, we have focused on data analysis and decision-making at the conclusion of a basket trial. A natural extension is to apply the proposed approach to basket trial design, incorporating interim analyses after prespecified numbers of enrollments. The proposed framework can be adapted to guide early stopping decisions for futility or efficacy while borrowing information across baskets. For example, after n_j^{int} patients have been enrolled in basket j , one may test $H_{0j}^{\text{int}} : \theta_j \leq \theta_{0j}^{\text{int}}$ versus $H_{1j}^{\text{int}} : \theta_j > \theta_{0j}^{\text{int}}$, where θ_{0j}^{int} need not be identical to θ_{0j} and may incorporate a target response rate as in Berry et al. (2013). If H_{0j}^{int} is not rejected, patient accrual in basket j may be stopped for futility. The proposed approach can be applied to such interim hypothesis testing, with tuning parameters g_1^{int} , g_2^{int} , and c^{int} calibrated to achieve desirable operating characteristics.

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Supplementary materials

Supplementary material is available at [Biometrics](#) online.

Web Appendices referenced in Sections 1, 2, 4, and 5 and data and code to reproduce the results in Sections 4 and 5 are available with this paper at the Biometrics website on Oxford Academic.

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Conflict of interest

None declared.

Data availability

The data that support the findings in this paper are available in the article and in its online supplementary material.

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